

Free-First SaaS Build Plan for BusinessOS AI and an Accounting System as a Service

Executive summary

This report treats **BusinessOS AI** as the prior in-conversation brief: a multi-module operating system for service businesses that unifies CRM, projects, proposals, invoicing, client collaboration, knowledge retrieval, automations, analytics, and an AI copilot. It treats the second product as a **TallyPrime-like multi-tenant accounting SaaS** focused on core bookkeeping and compliance workflows rather than the full breadth of desktop ERP features. Team size, budget, primary country, payment rails, bank-feed providers, deployment regions, data-residency requirements, and exact AI model vendors are **unspecified**. That matters because those unknowns affect pricing, compliance scope, and the right tenancy model.

The two products should not be built the same way. BusinessOS AI is well suited to a **pooled modular monolith** with tenant-scoped data and PostgreSQL Row-Level Security, plus an optional bridge path for premium isolated tenants. The accounting SaaS should start with a **bridge model**: a shared control plane for auth, onboarding, billing, and support, but stronger isolation for ledger data, ideally **schema-per-tenant or database-per-tenant for the accounting core**. This recommendation follows the SaaS tenancy patterns documented by Amazon Web Services ¹ and the database-per-tenant guidance published by Neon ², while relying on PostgreSQL RLS for any pooled tables that remain shared. ³

The economic picture is favorable for prototyping and early MVP work. A real prototype can run at effectively **\$0 platform spend** if it stays inside free tiers such as Cloudflare Pages, GitHub Free, Neon Free, Supabase Free, Resend Free, PostHog Free, and Figma Starter. However, once background jobs, long-lived workers, PDF rendering, bank imports, tax reports, and production-grade observability become mandatory, “free-only” usually stops being realistic. The practical interpretation of “free/open-source first” is therefore: **use free managed tiers where they are generous, and self-host only the pieces that create strategic leverage or materially reduce variable cost.** ⁴

The effort profiles are also asymmetric. A focused 60-day BusinessOS AI MVP is achievable in roughly **32–40 person-weeks** if AI scope is disciplined and accounting remains light. A credible accounting MVP with double-entry integrity, reporting, tax scaffolding, and reconciliation is more realistically **40–52 person-weeks**, even before multi-country tax localization. The reason is not UI complexity; it is the need for immutable journal logic, strong auditability, period locks, reconciliation, and localization boundaries. TallyPrime’s own workflows around chart of accounts, voucher types, payments, bank reconciliation, GST reconciliation, and e-Way Bill handling illustrate how quickly accounting scope expands once compliance enters the product surface. ⁵

Product	Recommended tenancy	MVP complexity	Estimated effort	Strategic conclusion
BusinessOS AI	Pool by default, RLS on shared data, bridge option for premium isolated tenants ⁶	Medium	32–40 person-weeks	Faster time to market, easier to differentiate with workflows and AI
Accounting SaaS	Bridge from day one: shared control plane plus schema-per-tenant or database-per-tenant ledger core ³	High	40–52 person-weeks	Stickier and more defensible, but correctness and localization dominate schedule risk

Product scope and core modules

The most important scoping decision is to **avoid parity thinking**. BusinessOS AI should not try to become a full ERP in its first 60 days. The accounting SaaS should not try to match every TallyPrime module, especially payroll, inventory, manufacturing, and every tax jurisdiction. Start with the economic center of gravity of each product.

BusinessOS AI scope

For BusinessOS AI, the right MVP is a service-business work system: pipeline, contacts, work execution, proposals, invoices, documents, client portal, workflow automation, analytics, and a bounded AI assistant. The strongest differentiation is not raw AI chat; it is AI grounded in tenant documents, projects, contacts, proposals, and billing context. The prior brief also supports an opinionated, clean UI and a modular monolith approach, which aligns well with Next.js App Router and an API-led backend. Next.js explicitly supports App Router, server and client components, route handlers, deployment guidance, and authentication-oriented patterns, making it a strong front-end foundation for this kind of unified workspace. NestJS, likewise, is designed for scalable, maintainable, testable server-side applications and includes recipes for Swagger, CQRS, health checks, Prisma, and async context handling. ⁷

Module	What belongs in MVP	Keep out of MVP
Tenant control plane	Workspaces, memberships, plans, usage metering	Multi-region tenancy controls unless a customer demands it
CRM	Contacts, organizations, notes, pipeline stages, deals	Complex forecasting, commissions
Work management	Projects, tasks, assignees, due dates, time logs	Resource planning, Gantt-heavy PM
Documents and knowledge	File upload, foldering, versioning, search, chunking for AI retrieval	Enterprise DMS features, legal hold
Commercial ops	Proposals, contracts, invoices, payment status, reminders	Full accounting ledger, payroll

Module	What belongs in MVP	Keep out of MVP
Portal and comms	Client portal, comments, email notifications	Omnichannel support inbox unless required
AI layer	Search, drafting, summarization, task extraction, proposal drafting, copilot over tenant data	Unbounded autonomous agents, unsupervised external actions
Automations	Triggers, rules, job runs, webhook receivers	Deep marketplace of hundreds of connectors
Analytics	Usage dashboards, pipeline funnel, project health, billing metrics	Full BI warehouse from day one

Accounting SaaS scope

For the accounting product, the MVP should emulate the **core accounting spine** rather than the whole TallyPrime surface. TallyPrime's own documentation shows the minimum viable workflow cluster: chart of accounts, voucher-driven journal posting, payments and receipts, bank reconciliation, GST return activity tracking, GST reconciliation, and e-Way Bill/e-invoice-adjacent flows for India. ERPNext, Odoo, Akaunting, and LedgerSMB document the same enduring accounting primitives in different product shapes: chart of accounts, journal entry, taxes, payments, bank reconciliation, localizations, tax reports, balance sheet, P&L, general ledger, and trial balance. ⁸

A disciplined MVP should therefore cover **one country first**. If India is the primary market, build for **GST-first India** and postpone other fiscal localizations. If not India, choose one VAT jurisdiction and defer GST/e-Way/e-invoice specifics. Odoo's fiscal localization model and TallyPrime's GST-specific reconciliation and return tracking are clear evidence that tax jurisdiction is a major architecture variable, not a cosmetic setting. ⁹

Accounting workflow	MVP module	Key source-of-truth tables	Why it matters
Chart of accounts	Accounting setup	<code>companies</code> , <code>fiscal_years</code> , <code>periods</code> , <code>accounts</code> , <code>account_groups</code>	Every downstream report and posting depends on COA structure
Journal entries	Ledger engine	<code>journals</code> , <code>journal_entries</code> , <code>journal_lines</code> , <code>posting_batches</code>	Double-entry correctness and auditability begin here
Ledgers and balances	Read models	<code>account_balances_daily</code> , <code>general_ledger_lines</code> , <code>trial_balance_snapshots</code>	Reports should be derived from journal lines, not hand-maintained balances

Accounting workflow	MVP module	Key source-of-truth tables	Why it matters
Invoicing	AR/AP	sales_invoices , sales_invoice_lines , purchase_bills , credit_notes , debit_notes	Operational documents that generate ledger postings
Payments	Cash and settlement	payments , payment_allocations , bank_accounts , payment_methods	Settlement, partial allocation, and aging depend on this layer
Tax and filing	Tax engine	tax_codes , tax_rates , tax_rules , tax_jurisdiction_settings , tax_returns	Must be country-specific from the beginning
Reconciliation	Banking	bank_statements , bank_statement_lines , reconciliation_sessions , reconciliation_matches	High-value MVP feature for reducing accounting workload
Reporting	Financial reporting	Derived SQL/materialized views over journal and operational tables	P&L, balance sheet, cash flow, GST/VAT reports, AR/AP aging

Open-source accounting products to emulate and what to borrow

If the accounting system must be built **from scratch**, the smartest use of existing open-source products is as **reference implementations**, not necessarily as the permanent foundation.

Reference product	What to borrow	What to be careful about
ERPNext	Company-scoped chart of accounts, journal entry model, tax templates, reporting coverage, immutable ledger direction from v13 onward ¹⁰	Broad ERP surface can distort a SaaS MVP if copied wholesale
Akaunting	Small-business invoicing UX, client-facing simplicity, lightweight dashboards, SMB-facing flows ¹¹	Its chart-of-accounts and double-entry path is app-based in current docs, so it is weaker as the canonical ledger reference for a new accounting core ¹²
Odoo	Reconciliation flows, standard financial reports, tax reporting, fiscal localization concepts ¹³	Feature boundaries vary by edition and localization stack; verify exact module licensing before using it as a parity benchmark

Reference product	What to borrow	What to be careful about
LedgerSMB	Country-localized chart templates, strong ledger orientation, country-specific tax form thinking ¹⁴	UX and operational ergonomics are weaker references for a modern SaaS interface; reconciliation docs are less polished ¹⁵

Architecture and data models

The architectural principle that should govern both products is simple: **monolith first, isolation where risk justifies it, and asynchronous boundaries around non-transactional work**. That means one deployable application initially, but not one flat data model.

Recommended tenancy model

For pooled systems, PostgreSQL RLS is an important control because it restricts rows per-role, defaults to deny when enabled without policies, and supports command-specific policies. It is powerful enough for most BusinessOS AI tables if the application never connects with owner or `BYPASSRLS` roles. For Accounting, however, RLS alone is not enough as the sole boundary for ledger data because support workflows, customer restores, noisy-neighbor behavior, and jurisdiction-based storage tend to create operational pressure toward stronger tenant isolation. That is why the recommended accounting model is a **bridge**: pooled control-plane data plus more strongly isolated financial data. ¹⁶

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flowchart LR
    U[Users and staff] --> E[Edge and frontend]
    E --> W[Next.js web app]
    W --> A[API and domain services]
    A --> T[Auth and identity]
    A --> C[(Shared control-plane Postgres)]
    A --> S[Object storage]
    A --> J[Jobs and workflows]
    A --> M[Email and notifications]
    A --> O[Observability and analytics]
    A --> L[AI gateway and model layer]

    subgraph BusinessOS_AI_runtime [BusinessOS AI runtime]
        A --> B[(Pooled tenant app database with RLS)]
        B --> BK[Knowledge and search indexes]
    end

    end

    subgraph Accounting_SaaS_runtime [Accounting SaaS runtime]
        C --> D[(Per-tenant ledger DB or schema)]
        D --> R[(Read models and financial reporting)]
    end

```

```
D --> X[Tax and reconciliation services]
end
```

The architectural consequence is straightforward. In BusinessOS AI, `tenant_id` and RLS can be applied across almost every table except global metadata and public catalogs. In Accounting, keep the **control plane pooled** but move the **general ledger, journal lines, tax configuration, and reconciliation state** into a separate schema or database per tenant. This gives clean customer export/restore boundaries and keeps mistakes in finance materially harder to cross-contaminate. Neon's documented database-per-tenant patterns make this much more feasible than it used to be on classic managed Postgres. ¹⁷

BusinessOS AI data model and API surface

For BusinessOS AI, the data model should be operational, not abstract. The user experience depends on a unified graph of people, work, files, messages, and commercial documents.

Domain area	Key tables and fields	Public/internal API surface
Tenancy	<code>tenants(id, slug, plan, status)</code> , <code>users(id, email, name)</code> , <code>memberships(user_id, tenant_id, role)</code> , <code>usage_meters(tenant_id, metric, period, value)</code>	<code>/tenants</code> , <code>/memberships</code> , <code>/usage</code> , <code>/billing/webhooks</code>
CRM	<code>accounts</code> , <code>contacts</code> , <code>deals(stage, amount, owner_id, expected_close_at)</code> , <code>activities(type, related_id, due_at)</code>	<code>/accounts</code> , <code>/contacts</code> , <code>/deals</code> , <code>/activities</code>
Work	<code>projects</code> , <code>tasks(status, priority, assignee_id)</code> , <code>time_entries</code> , <code>comments</code> , <code>labels</code>	<code>/projects</code> , <code>/tasks</code> , <code>/time-entries</code> , <code>/comments</code>
Documents	<code>files(storage_key, mime, size)</code> , <code>documents(type, title, status)</code> , <code>document_versions</code> , <code>document_chunks</code> , <code>permissions</code>	<code>/files</code> , <code>/documents</code> , <code>/documents/search</code> , <code>/documents/share</code>
Commercial ops	<code>proposals</code> , <code>proposal_sections</code> , <code>invoices</code> , <code>invoice_lines</code> , <code>payment_requests</code> , <code>customer_portal_sessions</code>	<code>/proposals</code> , <code>/invoices</code> , <code>/invoice-reminders</code> , <code>/portal</code>
AI	<code>ai_threads</code> , <code>ai_messages</code> , <code>ai_runs(model, latency_ms, total_tokens, status)</code> , <code>prompt_versions</code> , <code>eval_results</code>	<code>/ai/threads</code> , <code>/ai/runs</code> , <code>/ai/search</code> , <code>/ai/feedback</code>
Automation	<code>workflow_defs</code> , <code>workflow_runs</code> , <code>jobs</code> , <code>webhook_endpoints</code> , <code>outbox_events</code>	<code>/workflows</code> , <code>/jobs</code> , <code>/webhooks</code> , <code>/events</code>

Domain area	Key tables and fields	Public/internal API surface
Analytics	product_events, dashboard_widgets, saved_reports	/analytics/events, /analytics/dashboards, /reports

The design objective is not to microservice each row cluster. It is to keep one application process while separating **transactional write models**, **async outbox**, and **analytics/event streams**. AI should never read “the whole tenant” through raw broad queries; it should read via documented scopes such as project, document set, client account, or proposal. That sharply reduces data leakage risk and improves explainability.

Accounting SaaS data model and API surface

The accounting system needs a much more rigid write path. The correct model is **append-only journal first, derived ledger and reports second**. TallyPrime’s voucher-centric approach and ERPNext’s journal-entry and immutable-ledger direction both point to the same operational truth: invoices, bills, and payments are business documents, but the general ledger remains the canonical financial record. ¹⁸

Module	Key tables and fields	API surface
Company and periods	companies(id, base_currency, tax_country), fiscal_years, periods(start_at, end_at, closed_at)	/companies, /fiscal-years, /periods, /periods/close
COA and dimensions	accounts(code, type, parent_id, normal_balance), dimensions(cost_center, project, branch)	/accounts, /account-groups, /dimensions
Journals and postings	journals(type), journal_entries(posted_at, source_type, source_id, reversal_of_id, prev_hash), journal_lines(account_id, debit, credit, txn_currency, base_amount, tax_code_id)	/journals, /journal-entries, /journal-entries/post
AR/AP documents	customers, vendors, sales_invoices, purchase_bills, document_lines, credit_notes, debit_notes	/customers, /vendors, /sales-invoices, /purchase-bills, /credit-notes
Payments and allocation	payments, payment_allocations(document_id, amount), bank_accounts, cash_accounts	/payments, /payment-allocations, /bank-accounts

Module	Key tables and fields	API surface
Tax engine	<code>tax_codes</code> , <code>tax_rates</code> , <code>tax_templates</code> , <code>tax_returns</code> , <code>jurisdiction_configs</code>	<code>/tax/codes</code> , <code>/tax/rates</code> , <code>/tax/returns</code> , <code>/tax/reconcile</code>
Reconciliation	<code>bank_statements</code> , <code>bank_statement_lines</code> , <code>reconciliation_runs</code> , <code>reconciliation_matches(confidence, status)</code>	<code>/bank-statements</code> , <code>/reconciliation/runs</code> , <code>/reconciliation/matches</code>
Reporting and audit	<code>audit_events</code> , <code>attachments(checksum)</code> , <code>materialized_trial_balance</code> , <code>aged_receivables</code> , <code>aged_payables</code> , <code>cash_flow</code>	<code>/reports/trial-balance</code> , <code>/reports/general-ledger</code> , <code>/reports/p-and-l</code> , <code>/audit-events</code>

The non-negotiable invariants for this accounting model are: debits equal credits at posting time; posted entries are never hard-edited, only reversed; periods can lock; tax configuration is versioned; and every financial document has an audit trail. Odoo's reporting model and LedgerSMB's tax-form orientation reinforce that these constraints are the difference between "invoicing software" and accounting software.

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Tooling and hosting tradeoffs

The tool choices below prioritize four things: low cash burn, clean modern UI, open operational escape hatches, and credible enterprise migration paths.

Recommended stack by layer

Layer	BusinessOS AI recommendation	Accounting SaaS recommendation	Strong free/open alternatives	Why this is the best starting point
Frontend	Next.js App Router + Tailwind + shadcn/ui	Same	Directus/Payload admin only for backoffice UX	Next.js App Router supports server/client components and route handlers; Tailwind is utility-first; shadcn/ui provides open-code components you can own and extend. 20

Layer	BusinessOS AI recommendation	Accounting SaaS recommendation	Strong free/open alternatives	Why this is the best starting point
Backend	NestJS modular monolith	NestJS modular monolith	Payload for content-heavy areas; Directus for admin panels and instant APIs	NestJS is built for scalable, maintainable server-side apps; Payload is a Next.js-native app framework/CMS; Directus auto-generates REST and GraphQL on SQL schemas. Do not put accounting posting logic into a no-code layer. ²¹
Database	PostgreSQL + Prisma in a pooled model with RLS	PostgreSQL + Prisma, but bridge isolation for ledger data	Supabase ²² for all-in-one MVP; Directus over SQL if admin-heavy	Prisma gives a type-safe ORM over Postgres; Supabase Free is generous for quick starts; PostgreSQL RLS is the right pooled control for tenant rows. ²³
Auth	Auth.js for app auth, sessions, and provider integration	Auth.js for MVP; Keycloak or ZITADEL once B2B SSO is needed	Keycloak, ZITADEL	Auth.js supports JWT or DB sessions and uses encrypted HttpOnly cookies for refresh-token patterns; Keycloak and ZITADEL add federation, SSO, MFA, SCIM-style enterprise paths, and multi-tenant identity models. ²⁴

Layer	BusinessOS AI recommendation	Accounting SaaS recommendation	Strong free/open alternatives	Why this is the best starting point
Object storage	SeaweedFS self-host or Supabase Storage	SeaweedFS self-host or managed object storage; avoid immature storage bets for the ledger stack	Cloudflare R2 if you later want managed object storage economics; not recommended as the only novel dependency for accounting MVP if ops are thin	SeaweedFS provides S3-compatible object storage with distributed architecture. I would not make MinIO the default for new greenfield builds in May 2026 because the public repo was archived on Apr 25, 2026. ²⁵
Queue and workflows	pg-boss first	pg-boss first; BullMQ only if Redis is already justified	BullMQ, Inngest	pg-boss uses PostgreSQL and <code>SKIP LOCKED</code> for reliable background processing, avoiding another infra component. BullMQ is excellent if Redis is present. Inngest is attractive for durable execution if you later prefer managed orchestration. ²⁶
Email	Resend ²⁷ for transactional mail	Resend for transactional; Brevo for higher daily send volume at low complexity	Brevo	Resend's free plan is clean for app-generated email and exposes a REST API over HTTPS; Brevo's free plan is generous for daily volume and contact storage. ²⁸

Layer	BusinessOS AI recommendation	Accounting SaaS recommendation	Strong free/open alternatives	Why this is the best starting point
Product analytics	PostHog Cloud Free or self-hosted PostHog	PostHog for app analytics; Metabase for finance/admin BI	Plausible CE, Matomo, Metabase	PostHog offers free monthly product quotas and self-hosting; Plausible CE and Matomo are strong privacy- and ownership-oriented alternatives; Metabase is excellent for embedded BI and internal finance dashboards. ²⁹
AI layer	LiteLLM + Langfuse + Ollama or vLLM	Same, but AI cannot own the posting path	Local-only models for privacy-sensitive flows	LiteLLM provides a unified gateway and budget controls; Langfuse adds LLM observability/evals; Ollama runs local models; vLLM exposes an OpenAI-compatible server for higher-throughput serving. ³⁰
Deployment	Cloudflare ³¹ Pages for frontend; workers/functions only where needed	Same for frontend; backend and jobs on a container host	Coolify, Dokku	Cloudflare Pages has a real free tier with 500 builds/month, 100 custom domains/project, unlimited sites, unlimited static requests, and unlimited bandwidth. Pages Functions share Workers quotas, so heavy dynamic apps eventually need a worker/container plan. ³²

Layer	BusinessOS AI recommendation	Accounting SaaS recommendation	Strong free/open alternatives	Why this is the best starting point
Monitoring	OpenTelemetry + Prometheus + Loki + Tempo + Grafana	Same, plus immutable audit events in app DB	GlitchTip for error tracking	OpenTelemetry is the vendor-neutral telemetry layer; Prometheus, Loki, Tempo, and Grafana cover metrics/logs/traces; GlitchTip is an open-source Sentry-compatible error platform. ³³
Design	Figma ³⁴ Starter	Same	Strict OSS design alternative not sufficiently verified in this research and therefore unspecified	Figma Starter is free and includes unlimited drafts with limited AI credits, which is enough for MVP UI systems and handoff. ³⁵
PM	GitHub ³⁶ Issues & Projects	Same	Plane CE, Taiga	GitHub Free includes Issues & Projects alongside unlimited repos and CI minutes; Plane CE and Taiga are credible self-hosted PM alternatives. ³⁷

Free-tier and hosting tradeoffs

The table below matters because “free” can mean two different things: **free managed usage** or **free software that still consumes server cost**.

Service	Highest-confidence free details in official docs	Practical use in these products
Cloudflare Pages	Free plan: 500 builds/month, 100 custom domains/project, unlimited sites, unlimited static requests, unlimited bandwidth; Pages Functions count against Workers quotas, with 100,000 daily requests across Workers/Pages Functions on the free plan. ³²	Ideal frontend/CDN for both products; do not overuse Pages Functions for heavy server work
GitHub Free	Unlimited public/private repos, 2,000 CI/CD minutes/month, 500 MB Packages storage, Issues & Projects. ³⁸	Best default for source, CI, and lightweight PM

Service	Highest-confidence free details in official docs	Practical use in these products
Neon Free	100 projects, 100 CU-hours monthly per project, 0.5 GB/project, autoscaling and branching included, 60K MAUs in Neon Auth. ³⁹	Excellent for BusinessOS AI pooled DB or accounting tenant DBs in early stages
Supabase Free	50,000 monthly active users, 500 MB database, 1 GB file storage, 5 GB egress, 5 GB cached egress. ⁴⁰	Strong all-in-one starting point for BusinessOS AI; more caution for an accounting-ledger core
Resend Free	3,000 emails/month, 100/day, 1 domain, 10,000 automation runs, 30-day retention. ⁴¹	Transactional email for both products in early stages
Brevo Free	300 daily emails, 100,000 contacts storage, 1 user. ⁴²	Better when daily marketing-ish volume is more important than developer-first email APIs
PostHog Free	1 project, 1-year retention, generous free tier; official site also advertises 1M analytics events/month and 5,000 recordings/month on the free tier. ⁴³	Strongest hosted analytics default if you want feature flags and session replay too
Figma Starter	Free, unlimited drafts, UI kits/templates, AI credits with starter limits. ³⁵	Sufficient for MVP design systems and flows

A practical reading of the above is this:

- **BusinessOS AI** can often stay inside free tiers longer because user value comes from workflow cohesion and scoped AI, not from thousands of expensive irreversible financial transactions.
- **Accounting SaaS** hits infrastructure reality sooner, because bank imports, PDF statements, ledger reports, reconciliation jobs, and tax processing create persistent worker and storage needs.
- If you want the cleanest low-ops path, use **Cloudflare Pages + Neon + Resend + PostHog + GitHub** for BusinessOS AI first.
- For Accounting, use **Cloudflare Pages + NestJS on a container host + Neon per-tenant DBs + pg-boss + Resend/PostHog**, and accept that the backend will likely outgrow “serverless-only” earlier.

Optional video stack if it becomes relevant later

Video is **not on the critical path** for either MVP. If BusinessOS AI later adds product onboarding, customer video hubs, or live events, the recommended free/open stack is: **tus/tusd** for resumable uploads, **FFmpeg** for transcoding, **Shaka Packager** for HLS/DASH packaging, **Video.js** or **hls.js** for playback, and **MediaMTX**, **SRS**, or **OvenMediaEngine** for live/low-latency streaming. ⁴⁴

Security, compliance, and AI safety

The baseline application-security controls for both products should be driven by OWASP ⁴⁵ ASVS and the current OWASP Top 10, not just framework defaults. ASVS exists precisely to define technical security

controls and secure-development requirements, while the OWASP Top 10 remains the mainstream risk taxonomy for web apps. For AI, the companion baseline should be the NIST ⁴⁶ AI RMF, whose core functions are Govern, Map, Measure, and Manage, combined with the OWASP GenAI/LLM risk guidance on prompt injection and related failures. ⁴⁷

For pooled data, PostgreSQL RLS must be treated as a **hard security boundary**, not a convenience filter. PostgreSQL documents that once RLS is enabled without policies, a default-deny behavior applies; it also documents that owners and `BYPASSRLS` roles can circumvent policies unless carefully configured. In practice, that means: use non-owner app roles, never let background workers connect as superuser, test RLS in CI, and prefer security-definer wrappers only in narrowly audited cases. ⁴⁸

For the accounting product specifically, the compliance posture must also include **accounting-control mechanics**:

Control family	BusinessOS AI	Accounting SaaS
Access control	Tenant-scoped RBAC + RLS for all shared modules	Tenant RBAC + stronger DB isolation for the ledger core + RLS on shared/admin tables
Change safety	Soft deletes, versioning, workflow audit logs	Immutable postings, reversals only, fiscal period locks, maker-checker approvals
Data governance	Export/delete flows, retention policies, audit logs	Export/delete plus legal retention, close-period preservation, signed exports
Secrets and integrations	Scoped API keys, signed webhooks, per-tenant provider tokens	Same, plus strict bank/tax-provider credential isolation
Observability	OTel traces, structured logs, run IDs	Same, plus posting batch IDs, reconciliation run IDs, tax submission traceability
Backup and recovery	PITR preferred, file checksum verification	PITR mandatory, tenant-level restore drills, journal-integrity validation after restore

AI safety controls

AI can add a lot of value in both products, but the allowed scope should differ sharply.

AI control	BusinessOS AI policy	Accounting SaaS policy
Retrieval scope	Tenant-only, document/project/deal-scoped retrieval	Tenant-only, workflow-scoped retrieval; no broad “entire books” queries by default
Prompt and model management	Version every system prompt and tool policy	Same, with stricter release approval

AI control	BusinessOS AI policy	Accounting SaaS policy
Tool permissions	AI may draft, summarize, classify, suggest tasks	AI may draft notes, classify documents, suggest reconciliations; it must not auto-post journals, alter taxes, release payments, or submit filings without human approval
Safety against injection	Strip/label untrusted document text; sandbox tools; disallow hidden instructions in imported content	Same, plus disallow any imported attachment from invoking ledger or tax actions
Measurement	Latency, token cost, success rate, human acceptance rate	Same, plus false-positive reconciliation rate and misclassification rate
Human oversight	User confirmation for outbound effects	Mandatory approval for every financial side effect

This control posture is consistent with the NIST AI RMF's govern/map/measure/manage cycle and with the OWASP GenAI guidance that highlights prompt injection as a primary contemporary risk. It also reflects the product reality that a bad AI draft in BusinessOS is usually recoverable, while a bad ledger posting or tax submission can create legal and audit consequences. ⁴⁹

MVP roadmap, pricing, and migration

The 60-day roadmap below assumes a small product team but converts effort into **person-weeks** because team size is **unspecified**. If you have fewer people, the roadmap stretches. If you have more people, do not parallelize the accounting core so aggressively that invariants fracture.

Prioritized 60-day MVP roadmap

Window	Product	Milestone and deliverables	Estimated effort
Days 1–10	Both	Foundations: tenant model, repo, CI, auth, role model, design system, observability skeleton, basic RLS tests	4 pw each
Days 11–20	BusinessOS AI	CRM + accounts/contacts/deals + projects/tasks + file upload	6 pw
Days 11–20	Accounting SaaS	Company setup, fiscal periods, chart of accounts, contacts, journals, currencies	6 pw
Days 21–30	BusinessOS AI	Proposals, invoices, client portal MVP, notifications, usage metering	6 pw
Days 21–30	Accounting SaaS	Posting engine, balanced journal validation, immutable posting/reversal flow, period lock	8 pw

Window	Product	Milestone and deliverables	Estimated effort
Days 31–40	BusinessOS AI	AI copilot v1: search, summarization, drafting, prompt/version registry, Langfuse instrumentation	7 pw
Days 31–40	Accounting SaaS	AR/AP documents, payments, allocations, credit/debit notes, aging bases	7 pw
Days 41–50	BusinessOS AI	Automations, scheduled jobs, webhook ingestion, dashboards, admin controls	5 pw
Days 41–50	Accounting SaaS	Tax engine v1 for one jurisdiction, bank statement import, reconciliation suggestions	9 pw
Days 51–60	BusinessOS AI	Hardening, QA, pricing gates, onboarding, release readiness	4–6 pw
Days 51–60	Accounting SaaS	Financial reports, exports, audit trail, hardening, release readiness	6–8 pw

Likely totals: BusinessOS AI **32–40 person-weeks**; Accounting SaaS **40–52 person-weeks** for one-country scope. Add a further **8–15 person-weeks** if you insist on multi-country tax localizations in the same window.

```

gantt
  title 60-day MVP roadmap
  dateFormat YYYY-MM-DD

  section BusinessOS AI
    Foundations, tenancy, auth, UI system      :a1, 2026-05-05, 10d
    CRM, projects, files                        :a2, after a1, 10d
    Proposals, invoicing, portal                :a3, after a2, 10d
    AI copilot v1                              :a4, after a3, 10d
    Automations, dashboards, admin             :a5, after a4, 10d
    Hardening and release                       :a6, after a5, 10d

  section Accounting SaaS
    Foundations, tenancy, auth, UI system      :b1, 2026-05-05, 10d
    Company setup, COA, journals, periods       :b2, after b1, 10d
    Posting engine and immutability             :b3, after b2, 10d
    AR/AP, payments, allocations                :b4, after b3, 10d
    Tax v1 and reconciliation                   :b5, after b4, 10d
    Reports, audit, hardening, release          :b6, after b5, 10d

```

Feature gating and pricing strategy

Because market, geography, support model, and competitive target are **unspecified**, the right output here is not a fake precise price list. It is a **pricing structure** tied to real cost drivers.

BusinessOS AI

Plan concept	Feature gates	Real cost drivers to meter
Trial	1 workspace, limited contacts/projects/files, capped AI usage, branded footer	Minimal support, tiny storage, low token usage
Starter	CRM + projects + proposals + invoices + basic portal	Seats, storage, outbound email, low AI token quota
Growth	More automations, analytics, client portal branding, integrations, expanded AI quotas	Token usage, workflow jobs, storage/egress, support load
Enterprise	SSO, audit export, custom retention, isolated tenancy option, custom AI/provider routing	Higher support burden, tenant isolation, custom compliance, premium uptime needs

BusinessOS monetization rule: charge on **seats + AI usage + automation volume + storage**, not just seats. Otherwise your most expensive customers become your least profitable.

Accounting SaaS

Plan concept	Feature gates	Real cost drivers to meter
Trial	1 company, 1 user, limited invoices and statements, no filing	Support and education cost
Standard	COA, journal, invoices, payments, core reports	Companies, documents, storage, support
Pro	Reconciliation, tax reports, API/webhooks, approvals, attachments	Statement lines, job volume, storage, support complexity
Enterprise	SSO, isolated DB, audit export, custom localization, premium support	Isolation, country-specific maintenance, support, governance

Accounting monetization rule: charge on **companies/entities + users + document or statement-line volume + compliance add-ons**, not AI. AI should be value-add, not the primary meter, because the product's durable value is correctness and compliance.

Migration path to microservices and enterprise features

Start with a modular monolith for both products. Split services only when the split reduces real risk or cost.

Extraction order	BusinessOS AI	Accounting SaaS
First	Notification/email worker	Ledger posting service boundary inside the monolith, with strict APIs and outbox
Second	AI gateway and evaluation service	Reconciliation service
Third	Integration/webhook hub	Tax/localization service
Fourth	Analytics/event warehouse	Reporting/warehouse service
Fifth	Search/document processing service	Document rendering/archive/export service
Sixth	Enterprise tenant isolation plane	Enterprise SSO, data residency, audit export, custom localization packs

The accounting-specific caveat is critical: **do not split journal posting across multiple services early**. Keep posting atomic in one transactional boundary. You may extract reconciliation, filing, imports, OCR, or analytics first, but the write-path into the ledger should remain singular until scale or audit constraints force a separate posting service.

Comparative analysis and recommendation

Dimension	BusinessOS AI	Accounting SaaS
Best reason to build	Faster differentiation through integrated workflows + AI assistance	Higher retention and mission-critical stickiness
Biggest technical risk	Scope sprawl and shallow AI that feels generic	Ledger correctness, reconciliation accuracy, tax/localization complexity
Best tenancy model	Pool with RLS, bridge for premium	Bridge from day one, with stronger isolation for ledger data
Best starting stack	Next.js, NestJS, Postgres, pg-boss, LiteLLM, Langfuse, PostHog	Next.js, NestJS, Postgres, pg-boss, Metabase/PostHog, one-country tax engine
Best monetization axis	Seats + AI + automation + storage	Companies + users + compliance/reconciliation volume
Hardest support burden	AI explainability and workflow setup	Accounting correctness, tax edge cases, migration/import support
Enterprise path	SSO, audit exports, isolated tenancy, custom AI routing	SSO, isolated DBs, audit exports, custom localizations, data residency
MVP effort	Lower	Higher

Dimension	BusinessOS AI	Accounting SaaS
Long-term defensibility	Moderate unless niche-specific	High if localization and operational accuracy are strong

If the objective is **the fastest credible SaaS launch**, BusinessOS AI should be built first. It tolerates pooled tenancy longer, benefits more from generous free tiers, and can reach usefulness without proving legal-grade correctness on day one. If the objective is **highest retention and strongest operational lock-in**, the accounting SaaS is the better long-term bet, but only if you narrow it to **one fiscal jurisdiction**, one migration path, and one strong bookkeeping workflow set.

The sharpest strategic recommendation is therefore this:

- Build **BusinessOS AI first** if your competitive edge is workflow orchestration, services operations, and AI-assisted execution.
- Build the **accounting SaaS first** only if you already have domain expertise in one country's bookkeeping and tax workflow, or a customer cohort explicitly asking for a modern web-native alternative.
- In either case, do **not** try to fuse both products into one launch. You can let BusinessOS AI stay "finance-light" at first, while the accounting product earns the right to exist as a separate ledger-grade system.

Open questions and limitations

The following details remain **unspecified**, and each one could materially change the design:

- Primary country and tax regime
- Whether India GST is the target localization
- Required bank-feed providers and statement formats
- Whether payment collection is only status-tracking or native payment acceptance
- Data residency and regional hosting needs
- Enterprise identity requirements such as SAML, SCIM, or customer-managed keys
- AI model providers, privacy posture, and whether local inference is required
- Whether document OCR is in scope for invoices, bills, or bank statements
- Whether the accounting product must include inventory, payroll, or manufacturing in MVP

Those are not minor implementation details. For the accounting product especially, they determine whether the roadmap stays inside a 60-day MVP envelope or turns into a much larger ERP program.

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